

Extendible Hashing

In extendible hashing, a hash table is stored in the main memory. and buckets are stored in the disk.

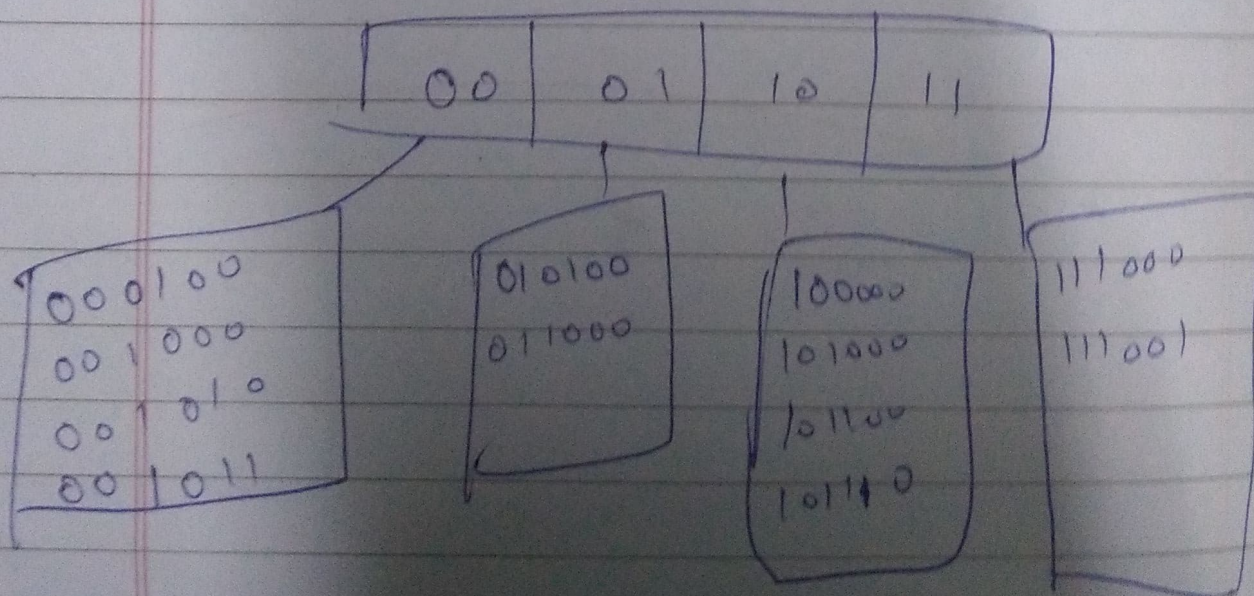
Root of the tree

00	01	10	11
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input = 6 digit number

Root of the tree contains 2 bits

$$\therefore \boxed{2^2 = 4 \text{ bits}}$$



Insert 100100

2 bit $\Rightarrow$ 3 bit $\therefore 2^3 = 8 \text{ bit}$

000	001	010	011	100	101	110	111
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100100

Random Probing

$$i \leftarrow K \bmod N + 1$$

$$\text{Keys} = 23, 45, 89$$

$$(23) = 23 \bmod 11 + 1$$

$$= 1 + 1 = \boxed{2}$$

$$(45) = 45 \bmod 11 + 1$$

$$= 1 + 1 = \boxed{2}$$

$$(89) = 89 \bmod 11 + 1$$

$$1 + 1 = \boxed{2}$$

$$i \leftarrow (i + m) \bmod h + 1$$

i = index

m = prime number

h = hash table size

$$i \leftarrow (2+5) \bmod 11 + 1$$

$$= 7 \bmod 11 + 1$$

$$= 7 + 1 = \boxed{8}$$

$$i \leftarrow (8+5) \bmod 11 + 1$$

$$= 13 \bmod 11 + 1$$

$$= 2 + 1 = \boxed{3}$$

$$i \leftarrow (3+5) \bmod 11 + 1$$

$$= 8 \bmod 11 + 1$$

$$= 8 + 1 = \boxed{9}$$

$$i \leftarrow (9+5) \bmod 11 + 1$$

$$= 14 \bmod 11 + 1$$

$$= 3 + 1 = \boxed{4}$$

~~$$i \leftarrow (9+5) \bmod 11 + 1$$~~

~~$$14 \bmod 11 + 1$$~~

~~$$3 + 1 = \boxed{4}$$~~

$$i \leftarrow (4+5) \bmod 11 + 1$$

$$= 9 \bmod 11 + 1$$

$$= 9 + 1 = \boxed{10}$$

$$i \leftarrow (10+5) \bmod 11 + 1$$

$$15 \bmod 11 + 1$$

$$4 + 1 = \boxed{5}$$

$$i \leftarrow (5+5) \bmod 11 + 1$$

$$10 \bmod 11 + 1$$

$$10 + 1 = \boxed{11}$$

$$i \leftarrow (11+5) \bmod 11 + 1$$

$$16 \bmod 11 + 1$$

$$5 + 1 = \boxed{6}$$

$$i \leftarrow (6+5) \bmod 11 + 1$$

$$11 \bmod 11 + 1$$

$$1 + 1 = \boxed{2}$$